CASE STUDY – II

SUPPLY, DEMAND & ELASTICITY

Why Did OPEC Fail to Keep the Price of Oil High?

Many of the most disruptive events for the world's economies over the past several decades have originated in the world market for oil. In the 1970s, members of the Organization of Petroleum Exporting Countries (OPEC) decided to raise the world price of oil to increase their incomes. These countries accomplished this goal by jointly reducing the amount of oil they supplied. From 1973 to 1974, the price of oil (adjusted for overall inflation) rose more than 50 percent. Then, a few years later, OPEC did the same thing again. From 1979 to 1981, the price of oil approximately doubled.

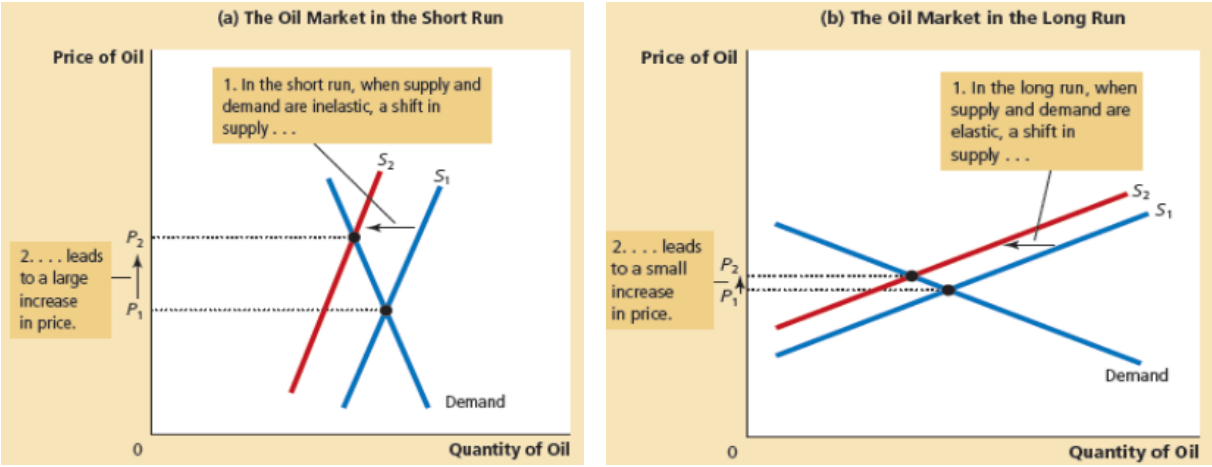
Yet OPEC found it difficult to maintain a high price. From 1982 to 1985, the price of oil steadily declined about 10 percent per year. Dissatisfaction and disarray soon prevailed among the OPEC countries. In 1986, cooperation among OPEC members completely broke down, and the price of oil plunged 45 percent. In 1990, the price of oil (adjusted for overall inflation) was back to where it began in 1970, and it stayed at that low level throughout most of the 1990s. (In the first decade of the 21st century, the price of oil fluctuated substantially once again, but the main driving force was changes in world demand rather than OPEC supply restrictions. Early in the decade, oil demand and prices spiked up, in part because of a large and rapidly growing Chinese economy. Prices plunged in 2008–2009 as the world economy fell into a deep recession and then started rising once again as the world economy started to recover.)

The OPEC episodes of the 1970s and 1980s show how supply and demand can behave differently in the short run and in the long run. In the short run, both the supply and demand for oil are relatively inelastic. Supply is inelastic because the quantity of known oil reserves and the capacity for oil extraction cannot be changed quickly. Demand is inelastic because buying habits do not respond immediately to changes in price. Thus, as panel (a) of Figure 2 shows, the short-run supply and demand curves are steep. When the supply of oil shifts from $S1$ to $S2$, the price increase from $P1$ to $P2$ is large.

The situation is very different in the long run. Over long periods of time, producers of oil outside OPEC respond to high prices by increasing oil exploration and by building new extraction capacity. Consumers respond with greater conservation, such as by replacing old inefficient cars with newer efficient ones. Thus, as panel (b) of Figure 2 shows, the long-run supply and demand curves are more elastic. In the long run, the shift in the supply curve from $S1$ to $S2$ causes a much smaller increase in the price.

This analysis shows why OPEC succeeded in maintaining a high price of oil only in the short run. When OPEC countries agreed to reduce their production of oil, they shifted the supply curve to the left. Even though each OPEC member sold less oil, the price rose by so much in the short run that OPEC incomes rose. By contrast, in the long run, when supply and demand is more elastic, the same reduction in supply, measured by the horizontal shift in the supply curve, caused a smaller increase in the price. Thus, OPEC's coordinated reduction in supply proved less profitable in the long run. The cartel learned that raising prices is easier in the

short run than in the long run.

Fig. (2): A Reduction in Supply in the World Market for Oil

When the supply of oil falls, the response depends on the time horizon. In the short run, supply and demand are relatively inelastic, as in panel (a). Thus, when the supply curve shifts from *S*1 to *S*2, the price rises substantially. By contrast, in the long run, supply and demand is relatively elastic, as in panel (b). In this case, the same size shift in the supply curve (*S*1 to *S*2) causes a smaller increase in the price.